Reinforcement Learning Lab

Lesson 2: MDP and Gym Environments

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Environment Setup

The first step for the setup of the laboratory environment is to update the repository and load the miniconda environment.

Safe Procedure

Always back up the previous lessons' solutions before executing the repository update.

- Update the repository of the lab:

```
cd RL-Lab
git stash
git pull
git stash pop
```

- Activate the *miniconda* environment:

```
conda activate rl-lab
```

First Tutorial

The README file on the repository (link) contains a simple tutorial on the basic libraries and tools we will use in the next lessons, in particular:

Numpy

NumPy is the fundamental package for scientific computing with Python. Think of it as a MATLAB-like environment for Python.

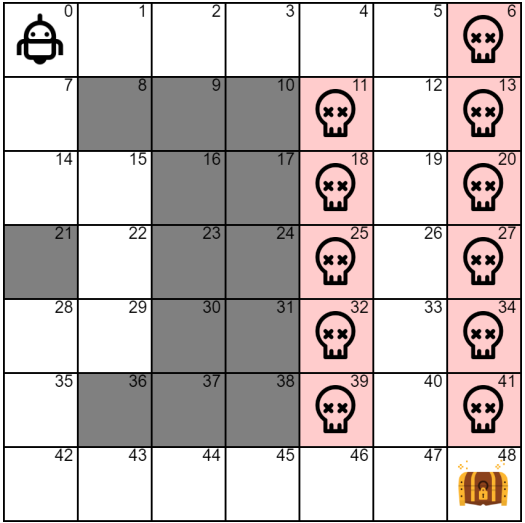
OpenAI Gym

Gym is a toolkit for developing and comparing reinforcement learning algorithms, a standardized set of tools and instructions to work with environments.

Keras

Keras is a high-level neural networks APIs that implements simple functions to create, train and modify neural networks.

Environment Description



The *DangerousGridWorld* environment. The goal for the robot is to reach the target (treasure) while avoiding the terminal states (death). Dark cells represent walls that can not be crossed.

All cells return a reward:

- **+5** for the target position (state 48)
- **-1** for the terminal cells (death)
- **-0.1** for each empty step

Today Assignment

In today's lesson, we experience gym environments. We implement a function to make random actions inside the given environment *DangerousGridWorld* generating a trajectory; after that, we implement an environment from scratch: *Recycling Robot*. In particular, the file to complete is:

```
RL-Lab/lessons/lesson_2_code.py
```

Inside the file, a *python function* and a *python class* are only partially implemented. The objective of this lesson is to complete them.

- **def random_dangerous_grid_world()**
- **class RecyclingRobot()**

Expected results can be found in:

```
RL-Lab/results/lesson_2_results.txt
```

Recycling Robot (a)

s	a	s'	$p(s' \mid s, a)$	$r(s, a, s')$
high	search	high	α	$r_{\texttt{search}}$
high	search	low	$1-\alpha$	$r_{\texttt{search}}$
low	search	high	$1-\beta$	-3
low	search	low	β	$r_{\texttt{search}}$
high	wait	high	1	$r_{\texttt{wait}}$
high	wait	low	0	-
low	wait	high	0	-
low	wait	low	1	$r_{\texttt{wait}}$
low	recharge	high	1	0
low	recharge	low	0	-

Figure: Description of the environment *Recycling Robot*, reporting the transition table from one state to the next and the reward for each action. A detailed description of the environment is available in the (book).

Recycling Robot (b)

Hint:

The first tutorial (environments) presents and describes all the functions and tools necessary to complete this first lesson.

Minimal Gym Environment

A minimal gym environment must implement the following functions: *init*, *reset*, *step*, and *render*. All these functions must be implemented in the class Recycling Robot. A complete example can be found (here).

DangerousGridWorld Disclaimer

DangerousGridWorld is not a *standard* gym environment. It implements different functions and simplified methods for learning purposes.